

VM-22 Implementation Research

FIA / SPIA / Fixed Annuities

VM-22 PBR: Implementation Challenges, Modeling, ALM, Market Impact

Research Date: 2026-05-19

Method: Perplexity AI Deep Research (Multi-Source Cross-Validation)

Language: Chinese / English Bilingual

概述

Overview

VM-22 (原则导向准备金框架 PBR) 自 2026年1月1日起生效, 取代传统的 CARVM 方式, 覆盖所有非可变年金 (固定指数年金 FIA、即期年金 SPIA、固定递延年金)。强制合规截止日为 2029年1月1日。

VM-22 (Principle-Based Reserving framework) took effect January 1, 2026, replacing the traditional CARVM approach for all non-variable annuities (Fixed Indexed Annuities, SPIAs, Fixed Deferred Annuities). Mandatory compliance deadline is January 1, 2029.

一、分产品影响

Product-Level Impact

FIA (固定指数年金) — 影响最大

- 准备金对股权类指数收益、利率路径、附加 rider (如 GLWB) 高度敏感
Reserves are highly sensitive to equity index returns, interest rate paths, and embedded riders (GLWB)
- 需要随机场景下的现金流建模, ALM 压力最大
Requires stochastic cashflow modeling - greatest ALM pressure
- 对冲策略设计直接影响准备金波动性和经济结果
Hedging strategy design directly impacts reserve volatility and economic outcomes

SPIA (即期年金) — 中等影响

- 保单持有人行为假设较简单, 但仍需比 legacy CARVM 更精细的现金流折现、资产匹配、经济场景测试
Policyholder behavior is simpler, but still requires more granular cashflow discounting and economic scenario testing
- 资产端与负债端的匹配度成为准备金关键变量
Asset-liability matching quality becomes a key reserve driver

固定递延年金 — 中等影响

- 复杂度介于 FIA 和 SPIA 之间
Complexity sits between FIA and SPIA
- 需要随机测试、资产分割、假设治理升级
Requires stochastic testing, asset segmentation, and assumption governance upgrades

二、建模工作核心挑战

Modeling Work Challenges

Dimension	Specific Challenge
Methodology Upgrade	From deterministic/formulaic reserves to multi-scenario stochastic projections
Exclusion Test	Support exclusion tests, deterministic reserves, and standard projections simultaneously
Dynamic Assumptions	Handle dynamic lapses, dynamic interest rates, negative rate scenarios
Nested Projections	Nested projections for pricing and asset adequacy dramatically increase computational load
Asset Modeling	Structured and alternative assets require more granular cashflow modeling
Infrastructure	The hardest part: rebuilding data pipelines, scenario engines, model governance, and reporting workflows

三、资产负债管理（ALM）考量

ALM Considerations

VM-22 下资产端对准备金结果的影响远大于 CARVM。

Under VM-22, assets have far greater impact on reserve outcomes than under CARVM:

Portfolio Construction: 按业务块的资产配置和再投资策略成为核心管理决策

Asset allocation and reinvestment strategy by block become central management decisions

Hedging: FIA 的对冲策略需要同时优化经济结果和法定准备金

FIA hedging strategies must optimize both economic results and statutory reserves

Data Integration: 负债模型与投资数据需更紧密集成

Liability models and investment data require tighter integration

Stress Scenarios: 错配风险、再投资约束在压力场景下直接影响准备金

Mismatch risk and reinvestment constraints directly impact reserves under stress scenarios

Structured Assets: MBS、CMBS 等含嵌入式期权的资产需更细致的处理

Assets with embedded options (MBS, CMBS) require more granular treatment

四、市场影响

Market Impact

Product Pricing: 准备金对产品设计和公司特定假设更敏感，部分 FIA 特征可能变得更昂贵

Reserves become more sensitive to product design and company-specific assumptions

Hedging Demand: 预计对冲工具需求和策略复杂度将上升

Expected increase in hedging instrument demand and strategy complexity

Asset Allocation: 推动更匹配的资产组合，减少流动性溢价依赖

Drives better-matched portfolios, reducing reliance on liquidity premium

Competitive Landscape: 建模和 ALM 能力强的公司在定价和风险管理上获得优势

Companies with stronger modeling and ALM capabilities gain competitive advantage

Efficiency Tools: 场景缩减、聚类分析、云计算需求上升

Demand for scenario reduction, clustering analysis, and cloud computing rising

五、精算工作量影响

Actuarial Workload Impact

Area	Impact
Annual Assumption Review	New mandatory requirement for annual review and update of assumptions
Scenario Testing	Significantly increased, hundreds of economic scenarios to maintain and run
Documentation & Governance	VM-31 reporting, sensitivity testing, exclusion test evidence required
Dual-Track Operations	Dual-track calculations required during the transition period
Staffing	Expected headcount growth across actuarial, IT, and risk management teams

六、实施时间线

Implementation Timeline

Year	Focus	Main Challenge
2024	Readiness, Interpretation, Gap Assessment	Understanding evolving guidance, identifying model gaps
2025	Build, Test, Governance Design	Model conversion, assumption process rebuild, control design
2026	Initial Implementation (New Business)	Prod-grade stochastic reserving + integrated ALM + reporting
2029	Mandatory Compliance Deadline	All blocks (including in-force) under PBR framework

关键结论

Key Takeaways

1. VM-22 不仅是准备金公式变更，而是年金保险公司的运营模式变革
1. VM-22 is not just a reserving change but an operating model transformation for annuity insurers
2. FIA 是战略影响最大的产品线，ALM 和对冲策略需要根本性重新设计
2. FIAs face the most strategic impact - ALM and hedging strategies need fundamental redesign
3. 建模基础设施升级是最大瓶颈，而非准备金公式本身
3. Modeling infrastructure upgrade is the biggest bottleneck, not the reserve formula
4. 2026 年是生产化元年，但存量业务的全面合规要到 2029 年
4. 2026 is Year 1 of production, but full in-force compliance is by 2029
5. 精算工作量显著上升，尤其是假设治理、VM-31 报告和双轨运行
5. Actuarial workload increases significantly in assumption governance, VM-31 reporting, dual-track ops

参考文献

References

- [1] PwC: Navigating VM-22 - Insights and Implementation Challenges
- [2] SOA: VM-22 Insights and Implementation Challenges (Sep 2025)
- [3] Milliman: VM-22 Readiness - Key Areas for Consideration
- [4] Milliman: VM-22 Hedging Strategies - Risk Mgmt and ALM for Fixed Index
- [5] Oliver Wyman: VM-22 Triggers A Strategic Reset (Feb 2026)
- [6] Aon: Fast and Flexible Technology to Navigate VM-22 (Jun 2025)
- [7] Am. Academy of Actuaries: VM-22 Retrospective Application (Jul 2025)
- [8] NAIC: Life Actuarial Task Force Materials (Spring 2026)
- [9] SOA: Emerging PBR Framework for Non-VA Annuities (Jun 2023)